Synthetic Proton/Neutron Reconstruction Study

B = 1.15 T, Beam Deflection = 3.0°

SMSimulator5.5 Analysis

March 8, 2026

Study Goal and Configuration

Goal: Evaluate reconstruction quality with synthetic beam input and focus on

$$\Delta P_x = P_x^{\mathrm{reco}} - P_x^{\mathrm{true}}.$$

Configuration

- Magnetic setting from macro: `run_protons_B115T_3.0deg.mac`
- Geometry/field macro: `geometry_B115T.mac`
- Target from macro: $(-12.488, 0.009, -1069.458)\,\mathrm{mm}$, angle 3.0°
- Dipole field map: `180703-1,15T-3000.table`, rotation 30°

Synthetic Input Dataset

Generator setup

- Proton momentum: $(x_p, 0, 627)\,\mathrm{MeV}/c$, $x_p \sim \mathcal{U}[-150, 150]$
- Neutron momentum: $(x_n, 0, 627)\,\mathrm{MeV}/c$, $x_n \sim \mathcal{U}[-150, 150]$
- Generated events: 20,000
- Reconstruction analysis sample: 3,000 events

Event statistics in analyzed sample

- Events with reconstructed PDC track: 2597 / 3000 (86.57%)
- Events with neutron TOF reconstruction: 994 / 3000 (33.13%)
- Proton fit sample cap: 600 events (Minuit/Three-point branches)
- Three-point fit convergence: $362/600 = 60.33\%$

PDC Reconstruction Process (Charged Proton)

Input

- `FragSimData` PDC hits (U/V layers for PDC1/PDC2).
- Geometry from `geometry_B115T.mac` (PDC positions/angles + target transform).

Reconstruction chain

- Convert U/V hits to CoM points in each chamber, with configured smearing.
- Build one straight `RecoTrack` with two points:

$$\vec{r}_1 = \text{PDC1 reco point}, \quad \vec{r}_2 = \text{PDC2 reco point}.$$

- Target back-tracing in field map ($B = 1.15\text{T}$):
  - Method A: **TMinuit** RK-based target momentum fit.
  - Method B: **Three-point (free-parameter)** fit with 6D optimization $(x_0, y_0, z_0, p_{x0}, p_{y0}, p_{z0})$, constrained by target/PDC1/PDC2 distances.
- Validation observable:

$$\Delta P_x = P_x^{\text{reco}} - P_x^{\text{true}}.$$

NN Objective and Scope

Objective

- Reconstruct target momentum directly:

$$(\hat{p}_x, \hat{p}_y, \hat{p}_z) = f(\text{PDC two-point track}).$$

- This branch predicts momentum components, not $B\rho$, and does not use matrix inversion.

Scope in this revision

- Offline ML pipeline only (dataset/training/inference/evaluation scripts).
- Not yet integrated into the online C++ auto-chain backend.

Dataset Construction

Sources

- Simulation tree branches: `FragSimData` and `beam`.
- Geometry/coordinate transform from selected macro.

Features and labels

- Input $X \in \mathbb{R}^6$: `pdc1_x,y,z`, `pdc2_x,y,z`.
- Target $Y \in \mathbb{R}^3$: $(p_x^{true}, p_y^{true}, p_z^{true})$ of proton truth momentum in target frame.

Filtering and outputs

- Keep events with proton truth, valid reconstructed PDC track, and finite feature/label values.
- Output files: `dataset_train.root` and `dataset_train.csv`.
- Builder script: `scripts/reconstruction/nn_target_momentum/build_dataset.C`.

Train/Validation/Test Split and Model

Split policy

- Formal campaign uses three independent simulation runs (train/val/test).
- Seeds: train 20260301, val 20260302, test 20260303.
- `train_mlp.py` consumes external split files (`--train-csv --val-csv --test-csv`).

Model and training

- MLP regressor: $6 \to 128 \to 128 \to 64 \to 3$ (ReLU).
- Input z-score normalization from training split statistics.
- Optimizer/loss: Adam + MSE with early stopping.
- Training script: `scripts/reconstruction/nn_target_momentum/train_mlp.py`.

Evaluation and Current Limitations

Where to read official hold-out metrics

- Hold-out metrics are reported from dedicated val/test datasets.
- Files: `predictions/metrics_val.json` and `predictions/metrics_test.json`.
- Training metadata also stores `val_metrics` and `test_metrics` in `model_meta.json`.

Generalization note

- Inputs currently include only PDC two-point coordinates.
- Beam/field configuration is learned implicitly from the training data distribution.
- Validity remains configuration-dependent; retraining is required when geometry/field changes.
- Effective dataset size is limited by PDC track availability after reconstruction cuts.

QMD-Window Retraining Test (New)

New campaign setup

- Geometry: `geometry_B115T_pdcOptimized_20260227.mac`, target 3°.
- Sampling narrowed to $p_\perp \leq 100$ MeV/c and $p_z \in [560, 680]$ MeV/c.
- Split size unchanged: train/val/test $= 200000/30000/30000$.

Baseline vs new test

Metric	Baseline (Wide)	QMD-window (New)
Train kept fraction	23.61%	44.48%
Val kept fraction	23.14%	44.16%
Test kept fraction	23.44%	45.02%
Val $\lvert\vec{p}\rvert$ RMSE (MeV/c)	9.19	6.01
Test $\lvert\vec{p}\rvert$ RMSE (MeV/c)	9.84	5.88

Conclusion: Narrowing to qmd-window improves both usable sample rate and hold-out momentum accuracy.

NN Optimisation Progress After QMD-Window

Configuration	Val $pxpy$ RMSE mean	Test $\lvert\vec{p}\rvert$ RMSE
QMD-window baseline	13.0186	5.8798
Best symmetric weighted+zscore	12.8172	5.3440
Best current asymmetric run	12.8027	5.2488

Interpretation

- Loss reweighting plus target normalization continues to improve hold-out metrics after the qmd-window gain.
- The current best result comes from asymmetric p_x-heavier weighting.

NN Current Status

Current training leader

- Run: `asym_20260307_120933_px431_zscore_256_256_128`
- Loss setup: manual weighting $4, 3, 1$ with target z-score normalization.
- Network width: $6 \to 256 \to 256 \to 128 \to 3$.

What we learned

- Asymmetric p_x-heavier weighting gives the best *pxpy* RMSE mean so far.
- Width increase alone gives only marginal gain after normalization is enabled.
- Optimisation is helping, but this is still an incremental campaign rather than a closed final model-selection step.

NEBULA Reconstruction Process (Neutron TOF)

Input

- `NEBULAPla` hit array (position, time, deposited energy).
- Target point from loaded geometry macro.

Reconstruction chain

- Time-window hit clustering (10 ns) and energy threshold cut (1 MeV).
- Pick best neutron candidate (highest reconstructed energy cluster).
- Flight vector:

$$\vec{L} = \vec{r}_{\mathrm{hit}} - \vec{r}_{\mathrm{target}}, \quad L = |\vec{L}|.$$

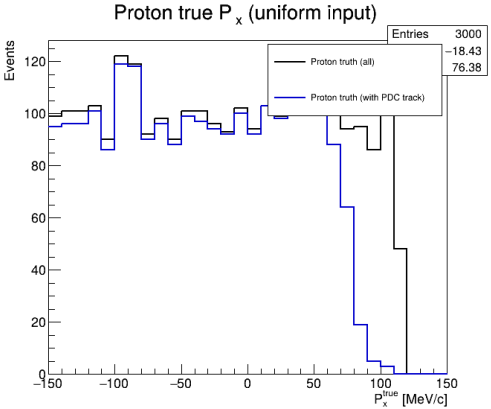
- TOF momentum:

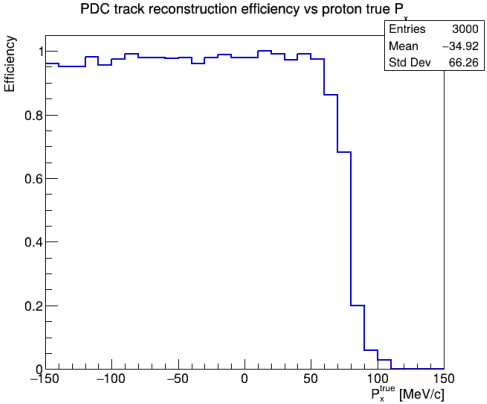
$$\beta = \frac{L/t}{c}, \quad p = \gamma m_n \beta c, \quad \vec{p} = p \cdot \frac{\vec{L}}{L}.$$

- Validation observable:

$$\Delta P_x = P_x^{\mathrm{reco}} - P_x^{\mathrm{true}}.$$

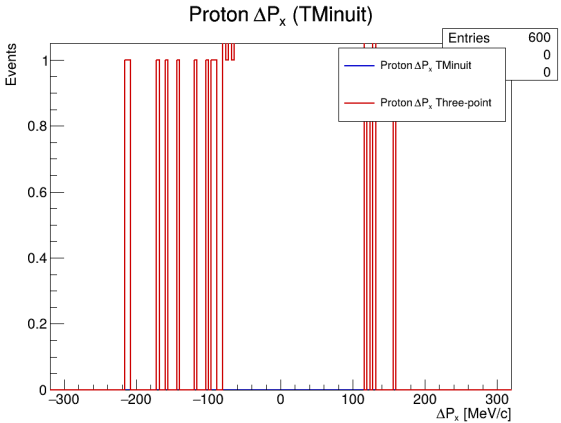
Uniform-Incidence Check: PDC Acceptance

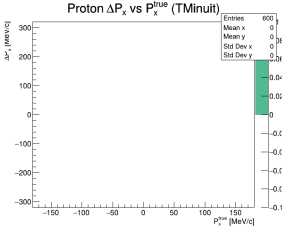


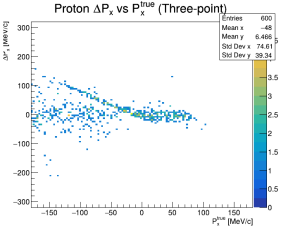


- For $P_x^{true} \in [-150, 80]$ MeV/c, PDC track efficiency is mostly $\sim 95\% - 99\%$.
- Efficiency drops at large positive P_x^{true}, indicating acceptance-edge loss.
- Global PDC track efficiency: $2597/3000 = 86.57\%$.

Proton ΔP_x: Key Result

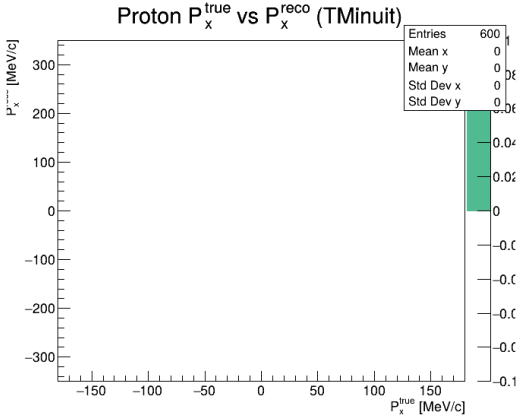


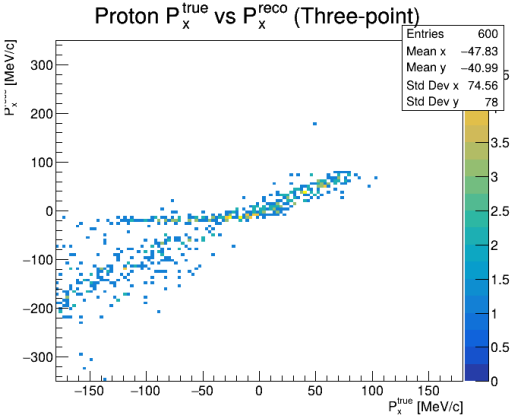




Method	Count	Mean(ΔP_x)	RMS(ΔP_x)
TMinuit	600	-517.086 MeV/c	60.603 MeV/c
Three-point (fwd)	600	+6.100 MeV/c	39.549 MeV/c

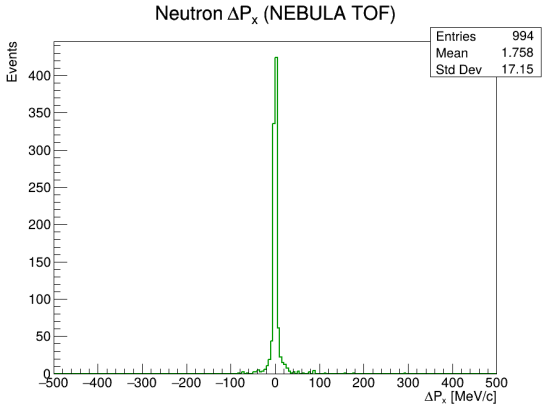
Proton P_x^{true} vs P_x^{reco}

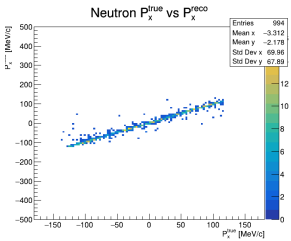


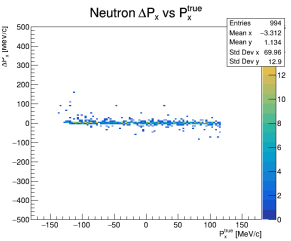


With $(0, 0, 627)$ initialization and priors, free three-point gives near-zero mean bias and moderate spread; Minuit keeps a large negative bias.

Neutron ΔP_x and 3D Momentum







Neutron TOF reconstruction summary (994 events):

$$\text{Mean}(\Delta P_x) = +1.758\ \mathrm{MeV}/c, \quad \text{RMS}(\Delta P_x) = 17.152\ \mathrm{MeV}/c.$$

Conclusions and Next Actions

Conclusions

- Uniform-incidence proton input is not reconstructed uniformly at high $+P_x$ due to PDC acceptance edge.
- Neutron TOF branch remains stable: Mean(ΔP_x) $= +1.758$ MeV/c, RMS $= 17.152$ MeV/c (994 events).
- Proton target back-tracing is method-dependent: Minuit has large negative bias; free three-point reaches Mean(ΔP_x) $= +6.180$ MeV/c, RMS $= 39.540$ MeV/c with 60.33% convergence.

Immediate next actions

- Keep three-point priors and tune convergence from 60% toward $> 90\%$.
- Fix two-point Minuit sign/bias issue (field/charge/direction convention check).
- Audit PDC acceptance edge ($+P_x$) with chamber-level hit occupancy.